# A CONDITIONAL SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON CANDIDATE

ANONYMOUS REPOSITORY AUDIT

Abstract. We study a proposed trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic Burau scalar $-59072$ in the grading convention corresponding to quantum degree 494. An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. We give an AR-side product-annulus model and formulate the candidate-specific Hattori and three-handle comparisons needed to promote the calculation to a smooth four-manifold invariant. Conditional on those comparisons, the resulting degree-494 class obstructs a diffeomorphism to $S^4$. The paper identifies three unresolved joins: the embedded detector collar, the actual MWW coefficient naturality squares, and a sphere replacement relative to the detector ball. The Lean development checks the finite and quotient algebra; it does not construct these geometric inputs or formalize four-dimensional differential topology.

Contents



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## 1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to $S^4$ is diffeomorphic to $S^4$. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to $S^4$ [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [12], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [15]. Manolescu and Neithalath gave a two-handlebody description [13], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [14]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of $S^4$, which is concentrated in bidegree zero.

We investigate the public finite class for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data. These ledgers do not by themselves supply an embedded framed Kirby presentation, an action-compatible coefficient comparison, or a sphere replacement fixed near the detector. These are retained explicitly as hypotheses.

**Contributions.** The main contributions are as follows.

(1) We give a public AR-side product-annulus model and two formal geometric cancellations, while isolating the missing embedding of the six-sweep detector collar in the actual reduced handlebody.
(2) We identify the local checked R-matrix with the public Burau blocks and prove the abstract divided beta/psi algebra, while retaining the candidate-specific Hattori and MWW naturality squares as a comparison gap.
(3) We explain how relative uniqueness of complete sphere systems and MWW's intrinsic local module action would yield the three-handle closure, and isolate the additional fixed-detector statement which is not proved by the cited theorem.
(4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance and are not replaced by narrative JSON fields or terminal status strings.

## 2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold $\Sigma_A^\varepsilon$, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that $\Sigma_A^\varepsilon$ is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction $\Sigma_A^\varepsilon$ is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [12]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

## 3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels $X$ and $S^4$, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

**Theorem 3.1** (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces $W_0, W_1, W_2, W_3$, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

*and linear isomorphisms*

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

*such that*

$$\ell_0(x_0) = -59072, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

*Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \to S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then $X$ is not diffeomorphic to $S^4$.*

*Proof.* Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of $F$ and $T$ implies $q_{12}(q_{01}(x_0)) = 0$. Applying $\ell_2$ and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If $X$ were diffeomorphic to $S^4$, the induced isomorphism would send $v$ to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. $\square$

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in the repository. We name the candidate-specific assumptions needed to instantiate them.

**Hypothesis 3.2** (Embedded candidate presentation, P0)**.** The public AR product witness is a labelled and framed handle presentation of $\Sigma_A^0$. Two product cancellations give the registered genus-two attaching link. A specified boundary 3-ball $B$ contains an actual six-sweep detector collar inducing the public 44- and 88-strand braids on all oriented cable states. The presentation has three 3-handles followed by one final 4-handle.

**Hypothesis 3.3** (Coefficient comparison, C)**.** The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from $B$. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is $-59072$ in the public normalization; all strict pivotal sign variants are nonzero. The class lies in quantum degree 494.

**Hypothesis 3.4** (Relative three-handle closure, S)**.** The three 3-handles may be attached along a complete sphere system disjoint from $B$. For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.

**Hypothesis 3.5** (Final candidate identifications, P3)**.** MWW's three-/four-handle formulas identify the final quotient with $\mathcal{S}^2_{0,(0,494),0}(\Sigma_A^0;\mathbb{Q})$. The corresponding summand for $S^4$ is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover, $\det(A-I)=1$ makes $\Sigma_A^0$ a homotopy sphere.

**Theorem 3.6** (Conditional trace-73 theorem)**.** *Assume Hypotheses 3.2–3.5. Then the resulting candidate satisfies*

$$\Sigma_A^0 \simeq S^4 \quad \textit{and} \quad \Sigma_A^0 \not\cong_{\mathrm{diff}} S^4.$$

*Proof of Theorem 3.6.* Hypotheses 3.2 and 3.5, and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Hypothesis 3.3 supplies the coefficient comparison, and Hypothesis 3.4 supplies the three-handle descent. They instantiate Theorem 3.1, whose selected class is nonzero in quantum degree 494. MWW's four-handle isomorphism preserves this degree, while the same degree of the standard $S^4$ module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to $S^4$. □

## 4. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. **Integral lattice calculations.** Besides (2), the repository records the proposed sphere coordinate columns

$$C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \qquad \det C = 1. \tag{6}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3;\mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. **The public Burau computation.** Let $\varepsilon = t-1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

1. parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
2. cable it to a 45,360-letter word on 88 strands;
3. apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
4. substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1+h)^{-2} - 1$;
5. extract the coefficient of $h^3$ after the specified source vector and covector are applied.

Denote the resulting scalar by $\delta_3^{\mathrm{Bur}}$. The independently reproduced public result is

$$\delta_3^{\mathrm{Bur}} = -59072 = (-2)^3 \cdot 7384. \tag{7}$$

The public input does not contain either decimal literal 7384 or $-59072$; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

**Proposition 4.1** (Exact scope of the computation)**.** *Equation (7) is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.3, however,*

$$\delta_3^{\mathrm{MWW}} = \delta_3^{\mathrm{Bur}} = -59072,$$

*up to the harmless global convention unit normalized in that theorem.*

*Proof.* The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Hypothesis 3.3; it does not follow from the finite script or from the local Burau-block calculation. $\square$

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

## 5. Lasagna modules and the intended obstruction

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold $W$ and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W;L)$ is generated by decorated surfaces in $W$, modulo local relations coming from $\mathrm{KhR}_N$ cobordism maps [15, 14].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where $W_1$ is a one-handlebody, $W_2$ adds two-handles along a framed link, $W_3$ adds three-handles along embedded spheres, and $W_4$ adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to $S^4$. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections record the proposed candidate-specific inputs and the points at which their identification remains unproved.

## 6. Compact product-ribbon presentation and the P0 gap

We record the public AR-side construction and then state the precise candidate-level embedding still required by Hypothesis 3.2. The construction does not recover the unavailable historical planar diagram.

6.1. **The actual AR link.** Use the coordinate-spine genus-three Heegaard splitting of $T^3$ from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a public Berkeley scan. Let $C_i$ be the three coordinate spine circles with common base vertex $Q$, and let $A_i$ be pairwise-disjoint narrow disk neighborhoods away from $Q$. Write $t$ for the mapping-torus one-handle. If

$\psi_A$ is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the $i$th mapping-torus two-handle is the embedded circle

$$C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i, \tag{8}$$

where $\lambda_i, \mu_i$ are the two arcs through $t$. The framing annulus is constructed simultaneously as

$$A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I). \tag{9}$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (8)–(9) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \qquad r_{yz} = [y, z], \qquad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle $h_{CS}$ goes geometrically once over $t$. Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \ldots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \qquad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \qquad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by $R$ transports the complete AR construction to the displayed matrix $A$. Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

6.2. **Return to the linear torus map.** Let $\phi_A$ be the linear torus map, straightened to the identity on the section ball, and choose an isotopy $\rho_u$, $u \in [-1, 1]$, from $\phi_A$ to $\psi_A$, stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A} \tag{10}$$

defines a diffeomorphism. Indeed, $g_{-1} = \mathrm{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (10) smooth, and the family $g_u^{-1}$ gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by

$F$ replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

6.3. **Words and normal fields.** Lift $C_i$ and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to $Ae_i$; it crosses the $j$th integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$m_i = t\phi_A(x_i)t^{-1}x_i^{-1}. \tag{11}$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1,1,1)$; the resulting strips have parallel boundary components, and linearity of $A$ preserves this property on their images [1, pp. 16–17]. Together with the $\lambda_i/\mu_i$ rectangles, these strips are the normal fields in (9). Thus no blackboard integer is substituted for a framing.

6.4. **The two cancellations.** Aitchison–Rubinstein identify $(t, h_{CS})$ as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over $h_{CS}$ along its product rectangle and cancel; this removes the $t, t^{-1}$ passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve $m_1$ becomes $zx^{-1}$ and has one geometric passage over the $x$ one-handle. Thus $(x, m_1)$ is another complementary pair. Sliding every other $x$ passage over $m_1$ along parallel product bands replaces $x$ by $z$ and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1,2,5,3,1)$, and the exact word projection is

$$|m_2| = 311, \quad [m_2]_{(y,z)} = (40, 269),$$
$$|m_3| = 1460, \quad [m_3]_{(y,z)} = (189, 1271).$$

For the committed $m_2$ word, a second linear and cyclic free reduction still has length 311; there is no adjacent inverse pair. Thus a separately reported length 309 uses a different convention and must be reconciled before comparison. This convention issue is not the load-bearing P0 gap. The transported $r_{zx}$ has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

6.5. **The actual detector ball.** Cut the remaining genus-two handlebody along its $y, z$ disk system. The selected $m_2/r_{xy}$ state has 42 $y$ passages from $m_2$ and two from $r_{xy}$. At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball $B$ containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of $B$. Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. The public verifier regenerates all 252 factors and obtains

$$|B_{44}| = 11340,$$

$$\mathrm{SHA256}(B_{44}) = \begin{array}{l}\texttt{7C2D2F792C2672221A76CAF08A71F560A}\\ \texttt{F0CB7654B4D537C00FAD00B16EFA187}\end{array}.$$

$$\mathrm{perm}(B_{44}) = 1, \qquad \mathrm{wr}(B_{44}) = 0.$$

The symbolic construction, source digest, matrix bridge, handle counts, cancellation data and collar labels are regenerated in `audit/t73_ar_product_witness.json`. Its digest is

AA9328EA3C7A53B937C97A02D611850D
0E8EEAF87C68756C04B013A486F6F112.

The source-PDF, regeneration and mutation checks are described in the repository README.

6.6. **Retained P0 comparison gap.** The preceding formulas define an AR-side handle model and an abstract pure braid in a standard punctured disk. They do not provide a labelled ambient isotopy or Kirby-move sequence placing the 44 selected passages in one specified framed 3-ball in the boundary of the actual reduced handlebody. In particular, the assertion at the start of the detector-ball construction that these passages have the required common regular neighborhood is an additional geometric input. The JSON witness records this assertion and its hash, but a SHA-256 digest proves only byte identity, not embeddedness or framing compatibility. Hypothesis 3.2 is exactly this missing embedding theorem.

There is a sharper provenance obstruction. The 252 public crossing rows come from the unavailable frozen planar diagram; the compact point-push script regenerates those rows but does not derive them from the AR component parametrizations or cancellation bands. Mapping-class realization and isotopy extension construct a chosen pure braid in a standard ball. They do not prove that the chosen braid is the relative-endpoint isotopy class of the 44 pre-existing AR passages: that relative class is exactly the braid-group datum to be proved. Thus the public 11340-letter word is not presently identified as an AR collar braid. In accordance with this dependency, the candidate-level comparisons in the next two sections remain proposals rather than closure arguments.

## 7. The coefficient-trace comparison

We describe the comparison required by Hypothesis 3.3. The local matrix, quotient algebra, and leading-order consequences are proved; the candidate-specific coefficient and all-state naturality maps are not.

7.1. **The product Hattori equivalence.** The compact words give

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

More is true than this count. In the cyclic $m_2$ word, every one of the 42 $y/Y$ events is followed immediately by a distinct $z/Z$ event; the same holds for the two $y/Y$ events of $r_{xy}$. The witness lists all 44 pairs. The corresponding one-edge intervals on the actual P0 product annuli are pairwise-disjoint subrectangles. Their removal leaves exactly 227 unpaired $z$ passages, closing to separate product circles between the two parallel annulus boundaries.

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \to P_{88}$, and let $F$ be the framed west-to-east tangle formed by the 44 rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \mathrm{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [14, Theorem 4.7]. The fixed product isotopy, pivotal tangle duality, and Künneth over $\mathbb{Q}$ give graded isomorphisms

(12)

$$H_{T,T'} : M_R(T, T') \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(FT, FT')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

Hypothesis 3.3 requires these maps to form a coefficient-bimodule isomorphism. The intended argument is that gluing $f : S \to T$ before the source link and then applying the fixed product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by $Ff$. The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface.

Apply the 227 disjoint counits to (12). This is a homogeneous coefficient morphism. The remaining coefficient $(T, T') \mapsto \mathrm{Hom}(FT, FT')$ is representable by the invertible tangle $F$ and hence is equivalent to the regular coefficient. The ordinary quotient statement, including both cyclic-relation spans, is kernel checked in `Smooth4PC/RepresentableCoefficient.lean`.

7.2. **Quantum trace and the endpoint map.** Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

Because (12) and the counits are homogeneous coefficient morphisms, they carry each such relation to the regular quantum vertical trace. BPW's canonical vertical-to-horizontal functor and the BHPW strict tangle/foam functor give

$$\mathrm{Sh}_q : q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow \mathrm{Hom}_{R_q}(E_{86}, E_{88}), \tag{13}$$

after the flat-base $\mathrm{qHH}_0$/Chern identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \qquad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1+h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo $h$ of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient $HH_0$; right exactness of tensor product suffices for this last assertion.

7.3. **Selected class and divided detector.** Let $U : P_{86} \to P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H^{-1}_{T_1,T_1}(\mathrm{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\mathrm{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1-q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart $P$ transports the operator, cup and cap simultaneously:

$$W \mapsto PWP^{-1}, \qquad u \mapsto Pu, \qquad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. BPW's cup/cap formulas then imply that the constant terms, after the same position change, are

$$u_0 = e_0 \pm e_5, \qquad \ell_0 = e^*_{87} \pm e^*_2.$$

All omitted factors are invertible q-monomials or positive-order $h$-corrections.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I)\,\mathrm{Sh}_h\,.$$

This is well defined on the quantum coefficient trace because (13) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]; equivalently, and also by direct evaluation of all 7,744 entries,

$$\rho_h(W) - I \in h^3\,\mathrm{End}(E_{88}).$$

Thus $D_h$ takes values in $h^3\mathbb{Q}[[h]]$. Division by $h^3$ is unique, and reduction modulo $h$ gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four possible strict pivotal signs give respectively

$$-53824, \quad -53760, \quad -59008, \quad -59072.$$

All are nonzero; the last is the public normalization. Hence

$$D_3([v_T]) = -59072 \neq 0 \tag{14}$$

in that normalization, without an unresolved sign premise.

7.4. **The complete two-handle cocone.** Only $D_3$ is asserted to descend. Filter the endpoint Temperley–Lieb category by through degree; composition cannot increase this degree. The selected cross-owner one-cup cell has through degree 86. The actual y-passage counts in owner order are

$$(42, 189, 2, 2, 0) \quad \text{for} \quad (m_2, m_3, r_{xy}, r_{yz}, r_{zx}).$$

The P0 normal fields provide arbitrarily many disjoint parallel levels. Taking parallel copies of the same product rectangles defines the Hattori shadow at every cable state. A beta move only changes their placement, and a psi move adds one further parallel negative/positive pair; hence these statewise shadows use one common geometric construction.

For beta, at every cable state apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Average the cubic row over the finite set of sign-preserving placements of the selected copies. If the detector uses $a_i^{\pm}$ copies among $k_i^{\pm}$ available copies of owner $i$, this orbit has

$$|\Omega_s| = \prod_i \binom{k_i^-}{a_i^-}\binom{k_i^+}{a_i^+}.$$

It is nonzero on every state on which the row is supported, so division is valid over $\mathbb{Q}$. Constant permutations reindex this average. After a placement is standardized, the remaining pure braid is $I + O(h)$; simultaneous transport of the operator, cup, and cap and $\rho(W) - I = O(h^3)$ show that it changes only terms of order at least four. Thus the cubic average is beta invariant. This argument does not assume that anti-parallel braid actions factor through a symmetric group.

For an undotted pair addition on any owner meeting the detector gate, the complete action-closed image has through degree at most 84 and vanishes in the degree-86 associated graded. The zero-gate owner is handled by the local core counits. For dotted and undotted additions the whole-source identities are

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \qquad (\epsilon \otimes \epsilon)\Delta(X) = 1.$$

The dotted raw degree $+2$ is cancelled by the target cabled shift $-2$. For a pair addition, forgetting the distinguished new pair has constant-size fibers between the relevant placement orbits. Hence the factor $|\Omega_s|/|\Omega_{s+e_i}|$ converts the target sum to the source sum. These ratios telescope under successive additions, so the target dotted row pulls back to the source row at every level.

These statements are the abstract algebra kernel checked in `Smooth4PC/TransportedPairing.lean`, `Smooth4PC/CubicJet.lean`, and `Smooth4PC/AugmentationCocone.lean`. They prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

Consequently $D_3$ descends through every relation in MWW's Definition 3.1, and Theorem 3.2 identifies the quotient with the actual $W_2$ lasagna module. Equation (14) proves that the image of $[v_T]$ is nonzero.

Finally, the detector motion and cap are supported in $B$. A cobordism outside $B$ is sent by the symmetric-monoidal endpoint functor to an independent tensor factor. Therefore the divided $W_2$ row has the monoidality used in Section 8.

The finite pairing list, four sign evaluations, source dependencies, and mutation controls are regenerated in `audit/t73_c_comparison_witness.json`; its digest is

```
410B1C411F321DD83F8ACD76CF5C259D
F2EE341629B13669C2FE458005E5D922.
```

7.5. **Retained C comparison gap.** The paper has not constructed (12) as a family of actual chain or foam maps for the P0 cut link. The statement that the 44 subrectangles and 227 residual circles are exactly the MWW coefficient is a geometric identification, not a consequence of their counts. Likewise, the beta averages and psi equations above specify the correct algebraic shape but do not define, for every cable state, the source and target rows or prove that the actual MWW beta/psi maps induce those rows. The cited Lean files prove the quotient and transport lemmas after such maps are supplied; they do not identify the maps for this handlebody. Thus Section 7 does not discharge Hypothesis 3.3.

## 8. Three-handle sphere closure and the relative gap

We record the absolute sphere-system argument and the retained relative condition in Hypothesis 3.4. Remove the final 4-handle and call the remaining handlebody $W_3$. Its boundary is $S^3$. Reversing the three 3-handles attaches three three-dimensional 1-handles to $S^3$, and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of $W_3$.

Fix the detector ball $B$ from P0. Choose a small ball in the connected complement of the actual attaching system. Since any two orientation-preserving embedded 3-balls in a connected oriented 3-manifold are ambient isotopic, the actual system may first be isotoped off the fixed ball $B$. Choose the standard three nonseparating spheres in the complement of $B$. Their complement is connected and their classes form the standard basis of $H_2^{\mathrm{sph}}(\partial W_2)$. Horvat–Jabłonowski's Theorem 5.3 identifies this system with the actual attaching system up to permutation, 3–3 handle slides and isotopy [10]. The relative uniqueness lemma was added in arXiv version 3 and is absent from version 2. Applying the version 3 statement to $Q = \partial W_2 \setminus \operatorname{Int} B$ makes the isotopies fixed on $\partial B$. The additional boundary slides become tubings to the inessential sphere $\partial B$ after the ball is restored and are removed by a local isotopy. Thus the 3-handles may be represented by standard spheres without moving the detector.

MWW's intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let $A_0$ denote the essential sphere with one dot and $A_1$ the undotted sphere. The quotient relations are

$$A_0 = 1, \qquad A_1 = 0.$$

The symmetric-monoidal naturality required in C applies because the standard spheres lie outside $B$. For every element $v$ in the complete source,

$$\ell(vA_0) = \ell(v)\operatorname{ev}(\dot{S}^2) = \ell(v), \qquad \ell(vA_1) = \ell(v)\operatorname{ev}(S^2) = 0.$$

Conditional on the fixed-$B$ comparison, these are whole-source equations. Hence the detector annihilates the full relation submodule for each of the three standard spheres and descends through the iterated MWW coequalizer.

The owner lifts, the 32-step Nielsen ledger and the split-tree Lean lemmas are compatible optional coordinate models, but they are not needed for this relative proof.

8.1. **Retained S relative gap.** The version 3 move list permits boundary slides over $\partial B$. The paper has not supplied a relative isotopy theorem showing that all such slides can be removed while the detector collar and its framing remain fixed inside $B$. Nor has it bound the resulting hemisphere maps to the actual MWW source and target maps. These are precisely the additional assertions in Hypothesis 3.4.

## 9. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-$(0,0)$ isomorphism [14]. Under Hypothesis 3.2, this is the handle decomposition of $\Sigma_A^0$. Under Hypotheses 3.3 and 3.4, the specified selected class remains nonzero in

$$\mathcal{S}^2_{0,(0,494),0}(\Sigma_A^0; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$, so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki's Proposition 2.1 and (2) prove that the genuine Cappell–Shaneson manifold $\Sigma_A^0$ is a homotopy sphere [11]. Applying these general results proves Hypothesis 3.5 once the candidate-level joins P0–S are supplied. The published general theorems do not themselves construct those joins.

## 10. Formalization boundary

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\texttt{ExternalGeometry} \longrightarrow \texttt{CSExternalGeometry} \longrightarrow \big(\mathrm{HS}(X) \wedge \neg\,\mathrm{Diff}(X, S^4)\big).$$

It does not construct either input structure.

The finite files check:

- the matrix $A$, $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

`IsActualHHO`, `IsActualSphereMap`,
`IsActualMWWCoequalizer`, `IsActualFourHandle`.

These are predicate fields of the ambient universe rather than definitions of smooth topology. Sections 6–9 explain proposed instances but do not construct the unresolved P0/C/S bindings. It would be incorrect to call the Lean development a formalization of four-dimensional differential topology or an unconditional kernel proof.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core, not a discharge of the candidate-level structure fields.

## 11. Status table and reproducibility ledger

The table records the current allocation of every external layer. Discharged means that a public generator or primary theorem and an independently checkable argument are both supplied.

| Item | Verdict | Reason |
|---|---|---|
| P0 | Open | The AR link and formal cancellations are available, but the public 11340-letter braid is not derived as the relative-endpoint braid of an embedded framed detector ball in the reduced AR link. |
| P1/C | Open | The Hattori formula and beta/psi cocone have the correct abstract shape, but the actual candidate coefficient maps and all-state naturality squares are not supplied. |
| P2/E7 | Partial | HJ gives absolute and relative sphere-system tools; the fixed-detector conclusion is not derived from the theorem and the candidate move list. |
| P2/E10/S | Open | The actual MWW hemisphere maps and their identification with the monoidal detector are not constructed. |
| P3/E11 | Partial | MWW Proposition 3.4 supplies the general four-handle theorem, conditional on the preceding candidate identification. |
| P3/E12 | Discharged | MWW Corollary 3.5 gives the standard $S^4$ module concentrated in bidegree zero. |
| P3/E13 | Partial | Iwaki's determinant criterion applies after P0 identifies the candidate with $\Sigma^0_A$. |

The historical availability script reports the large files as absent. The compact witnesses reproduce finite projections and proposed maps but do not replace the missing candidate-level embedding and naturality proofs.

## 12. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 12, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [13, 14]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]. Section 7 records the algebraic shape of the required candidate-specific comparison, but does not construct the actual MWW chain/foam maps or prove their naturality for all cable states.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [16]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the absolute replacement problem to boundary, embeddedness, disjointness and homology-basis checks. Version 3 adds a relative uniqueness statement for complete sphere systems, but the present paper does not derive from it an isotopy and slide sequence fixed on the specified detector ball. The owner-lift and Nielsen ledgers are useful coordinates, not a substitute for that missing relative argument.

## 13. Reproducibility and review

The proved finite and formal layers can be replayed in the following order.

(1) Regenerate the public AR product witness, point-push, Hattori and owner-lift ledgers.
(2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
(3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
(4) Compare the proposed C and S constructions with the cited functoriality theorems and the MWW handle formulas. Record the missing candidate-specific maps rather than treating this audit as a proof.
(5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.

The historical large artifacts are unavailable for this replay. Independent review should check the public AR mapping-torus diffeomorphism, product framings and two cancellations, and should attempt to construct the three presently absent objects: the embedded framed detector collar, the candidate MWW coefficient/naturality maps, and a complete sphere replacement fixed near that collar.

## 14. Conclusion

The public calculation and formal algebra produce a robust nonzero cubic scalar and show exactly how such a scalar would obstruct standardness after the relevant geometric and functorial identifications. They do not yet establish those identifications. Theorem 3.6 is therefore a conditional theorem, not a counterexample to the smooth four-dimensional Poincare conjecture. Closing P0, C, and S requires new candidate-specific geometric and link-homological arguments.

## Appendix A. Retired historical objects

The following table records historical artifacts cited by earlier project notes. They are unavailable and supply no current input. The compact AR, coefficient-comparison, and relative-sphere ledgers recover finite coordinates but do not discharge the old P0/P1/P2 roles. The digests are retained for provenance.

| Priority | Object | SHA-256 |
|---|---|---|
| P0 | `cs_presentation.py` | B20BDFF6FBF2A0E629083E2E62DD1DDF<br>9172A4E8893311CE122FFC4FEB388DFA |
| P0 | `t73_eps0.erkmo.json` | 6DF4D7906ADB18BA2D3237D4B49F8985<br>A5B18D8C39FF522AD6498F2369803FCB |
| P0 | `t73_reduced_billiard.pd.json` | E6912A64457557469E5C691B4D57ABDB<br>BF4C45ADB05492777C574223D0C06F8A |
| P0 | `CUT_OBJECT.json` | 4E1C4564ED98F30EAEB0A1AEAAD95ACA<br>B5AD7B226C79B7D5BF3F259E8B56F541 |
| P0 | `BASE_INTERACTION_SUPPORT.json` | 2D9E4760F836D9E04E6E0024CAE03A3D<br>BB49A74FF5D129756F25450B30F10F7C |
| P0 | `NORMAL_FIELD_MOVIE_CERT.json` | B2E97EB0981036768B8F1C1D5EF16F8<br>817EC1A7B951B6B32DBA2F9211FFD725A |
| P0 | `T73_COLLAR_BRAID.json` | 13AED414447ED33057C99D340910B501<br>09983CF36874C14ECE0E6A2834A12C24 |
| P0 | `verify_t73_collar_braid.py` | 9D93A6CB320200A906F461EF3131127E<br>A40C80A9A7828128E6F2804808417056 |
| P1 | `ACTUAL_PD_CABLE_UNIT_CERT.json` | 7F3D3618D6A790A9B60EE8085B647AC2<br>AB742E1BC9C15841F1BEF015034217B5 |
| P1 | `PRODUCT_NORMAL_CHRISTOFFEL_`<br>`THXY_MOVIE.json` | DE1AC479699EC79DE76D4265993DE437<br>493A3AAA6CABB636F98998644BF3181C |
| P2 | `TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json` | EE620E6B085A5F9E1C73CFDD1AD04FC<br>0682CEC74DA3DBF8AFE70DD19C038E3A0 |
| P2 | `TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json` | 4D1B627C0343A1C464319704EAFADCA1<br>27902A2E4E90CF1C283004359B7ADC24 |
| P2 | `THXY_FULL_MACRO_P3FREE_HJ_CERT.json` | EABF67C0D1AE309F0281297710A283B8<br>ED5A11D88C8E810837932313F4C53227 |

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. A proof of P0 must construct or replace the actual embedded detector object; a proof of C must construct the actual coefficient/Hattori maps and their functorial role. A proof of S must additionally establish a sphere replacement fixed near the detector and identify the resulting hemisphere maps with the MWW coequalizer maps.

## Appendix B. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

| Lean field | Mathematical obligation |
|---|---|
| `x0`, `ell0`, `ell0_x0` | P1 comparison and the selected genuine MWW class. |
| `q01`, `ell1_comp_q01` | Complete two-handle beta/psi quotient and descent. |
| `q12`, `ell2_comp_q12` | Actual E7 spheres, MWW hemisphere maps, and complete coequalizer. |
| `transport`, `fourIso` | E11 applied to the same graded lasagna module. |
| `s4DegreeZero` | E12 after identifying $G$ with rational $N = 2$ lasagna. |
| `diffeomorphismEquiv` | Graded diffeomorphism invariance of that same object. |
| `matrixConditionsToHomotopySphere` | E13 for the manifold identified by P0. |

These are exactly the candidate-level obligations isolated in Sections 5–8. They have not been discharged. The Lean theorem consumes no stronger statement, but it still requires them as structure fields.

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